ABSTRACT OF THE DISCLOSURE

An apparatus and method for the physical transmission of massively parallel, spatially-encoded optical data between an optical emitter array and an optical sensor array. The invention describes two primary embodiments for a photon-native transmission medium that avoids signal conversion bottlenecks. The first embodiment is a flexible "image conduit" cable comprising a coherent bundle of millions of optical fibers, where the spatial arrangement of the fibers is preserved from end to end, directly transmitting an image frame. The second embodiment is a rigid "light-pipe" interconnect using a system of collimating and focusing lenses within a sealed housing to project the image frames over a free-space optical path. Both embodiments provide a physical layer capable of supporting multi-terabit or petabit-per-second data rates by maintaining the inherent parallelism of a frame-based optical data transmission system.